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Testing VPC Connectivity



Mohammed Dawoud Mota

```
}  
  
.no-js-shield {  
  width: 2.4rem;  
  height: 2.4rem;  
}  
  
.no-js-logo-image {  
  width: 2rem;  
  height: 2rem;  
}  
  
@media (max-width: 768px) {  
  .no-js-heading {  
    font-size: 2.5rem;  
  }  
  
  .no-js-description {  
    font-size: 1.125rem;  
    width: 100%;  
  }  
  
  .no-js-container {  
    padding: 0 1rem;  
  }  
}  
  
</style>  
<div class="no-js-overlay">  
  <div class="no-js-logo">  
      
  </div>  
  <div class="no-js-container">  
      
    <h1 class="no-js-heading">JavaScript is required, whaaaaaat!</h1>  
    <p class="no-js-description">  
      NextWork requires JavaScript to function properly. Please enable JavaScript or disable any script-blocking  
      tools like Brave Shields to use this site.  
    </p>  
  </div>  
</div>  
</noscript>  
/body>  
/html>[ec2-user@ip-10-0-0-93 ~]$
```



Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a virtual network in AWS that lets us launch resources in a secure and isolated environment, giving control over subnets, IPs, routing, and access.

How I used Amazon VPC in this project

I used Amazon VPC to host public and private subnets, launch EC2 instances, and test connectivity between them and to the internet securely.

One thing I didn't expect in this project was...

I didn't expect that the first ping between EC2 instances would fail, which showed me how important security group rules are for connectivity.

This project took me...

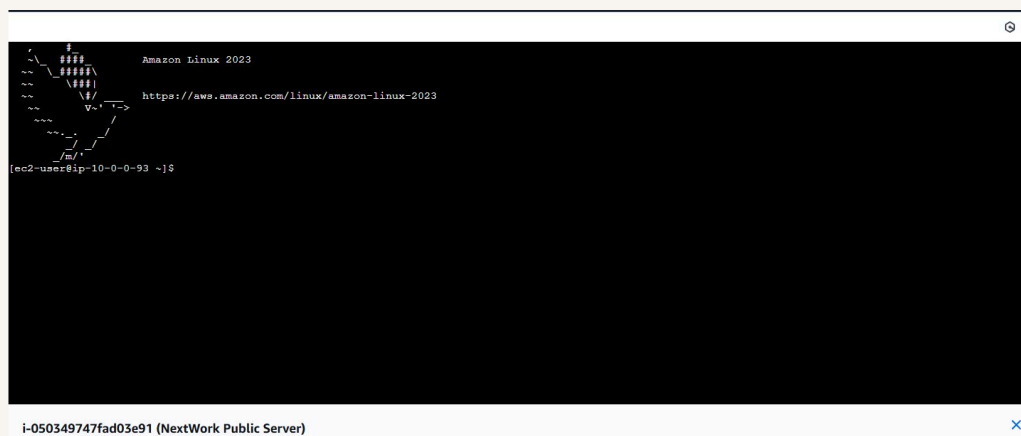
This project took me 30 minutes.



Connecting to an EC2 Instance

Connectivity means the ability of our resources, like EC2 instances and subnets, to communicate with each other and with the internet without any blocks or interruptions.

I was trying to connect to the public EC2 server in our VPC using the AWS Management Console.





EC2 Instance Connect

EC2 Instance Connect is a service that lets us securely connect to our EC2 instances using a browser or SSH without needing to manage key pairs manually.

My first attempt at getting direct access to my public server resulted in an error because the attached security group did not have a rule to allow SSH access from anywhere.

I fixed this error by going to the attached security group, clicking edit inbound settings, and adding a rule for SSH access from anywhere.



Failed to connect to your instance
Error establishing SSH connection to your instance. Try again later.



I ran the ping command followed by the private IP address of the other EC2 instance to test connectivity between them.

The first ping returned a timeout, which means the instances couldn't communicate due to security rules or network configuration issues.

```
#  
~\##### Amazon Linux 2023  
~~\#####\  
~~\###|  
~~\#/ https://aws.amazon.com/linux/amazon-linux-2023  
~~V~' '->  
~~~~/  
~~.-./-/  
~/m/' -/
```

[ec2-user@ip-10-0-0-93 ~]\$ ping 10.0.1.216
PING 10.0.1.216 (10.0.1.216) 56(84) bytes of data.



Connectivity to the Internet

Curl is a command-line tool used to send requests to a server and see its response, which helps test connectivity and access to websites or APIs.

I used curl to test if our EC2 instance could reach the internet and receive a response from a web server, confirming outbound connectivity.

Ping vs Curl

Ping checks if a server is reachable by sending packets, while curl requests data from a server and shows the response, testing both connectivity and service access.



Connectivity to the Internet

I ran curl followed by a public website URL to verify that our EC2 instance could access the internet and receive a response.

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